A Comparative Data Mining Framework Stock Price Forecasting on Indonesian IDX30 Using Statistical, Machine Learning, Deep Learning, Transformer, and Hybrid Models

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I. Introduction

Predicting stock prices remains a difficult task because market time series are typically non-stationary, highly volatile, and affected by both structured and unexpected events. This difficulty is amplified by both structured and unexpected events. This difficulty is amplified in emerging markets, where price behavior may respond strongly to local regulations, political context, and global macroeconomic movements. In Indonesia, the IDX30 index represents a set of highly liquid and lare-cap companies making it a meaningful benchmark for studying stock forecasting under realistic market dynamics.

From a data mining perspective, stock forecasting is not only a predictive problem but also a knowledge discovery process which has an objective of extracting patterns and relationships from historical records that can support decision-making. Rather than relying on a single algorithm, this work adopts a comparative approach across model families to understand how different assumptions and representations (linear time-series structure, engineered features, sequence learning, attention mechanisms, and interpretable decomposition) perform under the same data conditions.

The main research question addressed in this paper is: Which forecasting approach provides the most reliable next-day closing price predictions for IDX30 stocks under a standardized evaluation framework, and what trade-offs exist between accuracy and interpretability? To answer this, we design a unified data mining pipeline and evaluate five models like SARIMAX, XGBoost, LSTM, LightGBM and NeuralProphet using consistent data splits and forecasting metrics.

The key contributions of this paper are as follows: (1) the formulation of a consistent data mining workflow for IDX30 time-series forecasting that includes data understanding, feature engineering, and comparable evaluation; (2) an empirical comparison across five forecasting model families under identical test conditions; and (3) a discussion of interpretability and practical deployment considerations, enabling selection not only by metric performance but also by model transparency and operational suitability.

II. Theoretical Concept

A. Basic Concept

Stock prediction is frequently expressed as a supervised learning problem in which patterns that can be extrapolated to future price movements are inferred from historical market data. The time-ordered dataset of daily stock prices uses historical observations as predictors for a target variable, like the closing price of the following day. Raw price data is converted into useful representations that capture temporal dynamics like momentum and volatility through feature construction. Examples of these representations include lagged prices, trend measures, and technical indicators. Then, to strike a balance between model accuracy and generalization to new data, predictive models learn statistical relationships between features and future prices.

B. Models Used

For this research we decided to use these approaches/models for complex time series prediction:

1. Deep Learning (LSTM)

   Long Short-Term Memory (LSTM) networks are super popular because they can capture those long-term dependencies that traditional models miss. Studies have shown that LSTM models can beat older methods like ARIMA in forecasting stock trends. The drawback, though, is that LSTM is a total "black-box" model. Financial analysts need to explain why a prediction was made to stakeholders, and a black-box model can't do that transparently.

2. Hybrid Models (NeuralProphet)

   NeuralProphet is the new kid on the block, introduced in 2021. It combines deep learning with the structure and interpretability of the older Prophet model. This means it's designed to easily handle factors like market seasonality and external covariates (like inflation or commodity prices), which are key in markets like the IDX 30. Early tests showed

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NeuralProphet significantly improving forecast accuracy over the traditional Prophet model.

3. Statistical Method (SARIMAX)

   Traditional models like SARIMA and SARIMAX have been widely used to capture linear patterns and seasonality in financial time-series data. While they work well for short-term forecasting and stable markets, they struggle to handle nonlinear and highly volatile data such as stock prices from emerging markets like Indonesia's IDX30.

4. Machine Learning Models (XGBoost)

   Ensemble learning algorithms such as XGBoost have gained popularity for handling nonlinear relationships and feature-rich datasets. Prior research shows that XGBoost can outperform traditional statistical models when technical indicators and lagged features are included. However, these models do not naturally capture temporal dependencies unless they are engineered manually.

5. Gradient Boosting Models (LightGBM)

   Gradient boosting–based models, particularly Light Gradient Boosting Machine (LightGBM), have been widely adopted for time-series regression tasks due to their ability to model nonlinear relationships and feature interactions efficiently. Using a tree-based learning framework with histogram-based splitting, LightGBM maintains strong predictive performance while allowing for quick training and scalability on big datasets. LightGBM is frequently used with technical indicators and engineered lag features in financial forecasting, which enables it to capture short-term market dynamics without explicitly modeling sequential dependencies. LightGBM relies heavily on feature design to encode time-related information and has limited inherent temporal interpretability when compared to attention-based models, despite its strong empirical performance in stock prediction tasks.

III. Methods

A. Models Applied

We decided to use 5 models for this project - SARIMAX, XGBoost, LSTM, LightGBM, and Neural Prophets. SARIMAX is a classical statistical model that uses moving averages, volatility, calendar effects, and lagged market return to model linear relationships and autocorrelation. XGBoost serves as a machine learning model that utilizes lagged features, moving averages, crossovers, and volatility to handle nonlinear tabular data effectively. LSTM learns patterns in the order of closing prices, volume, moving averages, and momentum indicators over time. LightGBM enhances forecasting performance by utilizing gradient-boosted decision trees to model complex nonlinear relationships and interactions across engineered temporal and contextual features. Lastly, NeuralProphet combines trend, seasonality, and outside regressors like moving averages. This makes it easy to understand and iterate. These models work together to give us a complete way to look at and predict how stock prices will change.

B. Dataset Used

The dataset used in this project is a historical daily stock dataset for IDX30 equities (e.g., BBCA), formatted in a standardized OHLCV time-series structure where each row represents one trading day. The table includes Date as the time index and numeric market variables such as Open, High, Low, Close, and Volume. In this study, Close is defined as the main forecasting target (next-day closing price), while Open/High/Low/Volume are optionally used as supporting predictors depending on model capability (e.g., XGBoost and TFT can utilize multivariate inputs, while LSTM and NeuralProphet may use Close-only in a univariate setup).

This dataset requires minimal cleaning because it is already well-structured and consistent: timestamps follow a uniform datetime format, the data is naturally ordered by date, and it follows the expected trading-calendar behavior (weekends and market holidays are not present, which is normal rather than "missing data"). Additionally, the dataset exhibits negligible missingness and does not show common data integrity problems such as duplicated trading dates, mixed datatypes within columns, or invalid values (e.g., negative prices/volume). As a result, most preprocessing steps focus on model readiness (feature construction or scaling) rather than heavy data cleansing.

C. Preprocessing

Across all models, the core preprocessing steps are intentionally lightweight: (1) convert Date into a proper datetime type, (2) sort the data chronologically, (3) ensure price/volume fields are numeric, and (4) handle rare missing values using simple methods such as linear interpolation (and forward/backward fill as a safety step). Importantly, gaps due to non-trading days are not treated as errors, since they reflect the real market calendar. These steps are sufficient because the dataset is already clean and conforms to standard financial time-series formatting.

Model Spesific Preprocessing

1. SARIMAX

   SARIMAX uses the closing price series directly in its original scale, so no scaling is required. The preprocessing focuses on indexing the series by date, keeping chronological order, and removing or minimally filling rare missing values. The model is then trained using a chronological train–test split and produces forecasts with confidence intervals.

2. XGBoost

   XGBoost requires a supervised tabular format, so preprocessing mainly involves feature engineering rather than cleaning. After ensuring numeric integrity and filling rare missing values, the pipeline constructs lag-based and rolling-window features (e.g., lagged closes, returns, moving averages, rolling volatility, and calendar features). Rows that become invalid due to lag/rolling operations are removed using dropna() function.

3. LSTM

   LSTM requires fixed-length sequences, so preprocessing includes converting the close price series into sliding windows (lookback sequences).

Because neural networks are sensitive to scale, the close price is normalized using MinMax scaling, fitted only on the training set to avoid leakage. The data is then reshaped into the 3D tensor format needed by LSTM (samples × timesteps × features).

4. LightGBM

LightGBM requires a supervised tabular format, so preprocessing mainly involves feature engineering rather than heavy cleaning. After ensuring numeric integrity and filling rare missing values, the pipeline builds lag features (previous Close/Volume), rolling features (moving averages and rolling standard deviation for volatility), and calendar features (day-of-week/month). The target is set as the next-day Close by shifting the Close column forward, and rows that become invalid due to lag/rolling/shift operations are removed using dropna() function.

5. NeuralProphet

NeuralProphet uses the standard Prophet-style schema by renaming Date → ds and Close → y, and it operates on the time series on its original scale. Preprocessing is minimal: ensuring the correct column names, datetime formatting, chronological order, and light missing-value handling if required. NeuralProphet can then generate predictions that are directly comparable against actual values in the evaluation window.

D. *Evaluation Metrics*

To evaluate the performance of our models, we will use two widely recognized metrics in forecasting:

I. Root Mean Squared Error (RMSE): This metric calculates the square root of the average of the squared differences between predicted and actual values. It gives a relatively high weight to large errors, making it sensitive to outliers.

II. Mean Absolute Error (MAE): This metric measures the average magnitude of the errors in a set of predictions, without considering their direction. It is a more robust metric against outliers compared to RMSE.

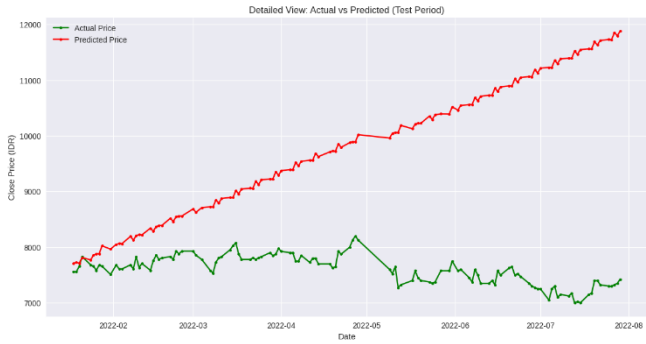
By comparing the RMSE, and MAE across the selected IDX 30 stocks, we aim to objectively assess the accuracy and reliability of the models for next-day stock price prediction in the Indonesian market.

IV. Results

A. *Experimental Results*

To evaluate the effectiveness of our forecasting framework, we tested the five selected models—SARIMAX, XGBoost, LSTM, LightGBM, and Neural Prophet—on the IDX30 dataset. The performance of each model was measured using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), as these metrics allow us to clearly see the average magnitude of the prediction errors. The visual comparison of these error rates is presented in **Fig. 1** below. A lower bar in the graph indicates a smaller error, meaning that the model's prediction was closer to the actual next-day closing price.

1. SARIMAX\

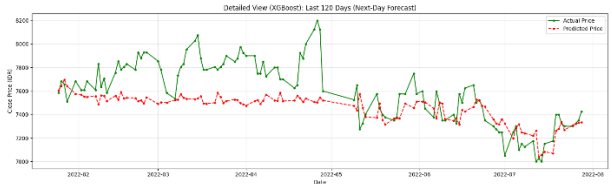


SARIMAX Performance Metrics (last 120 days):

MAE: 2,175.40 (28.62%)

RMSE: 2,588.19 (34.04%)

2. XGBoost

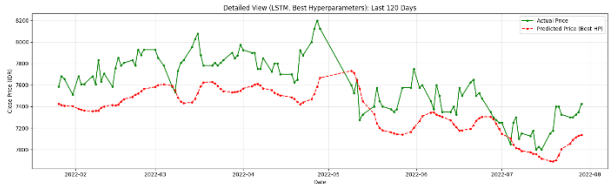


XGBoost Performance Metrics (Last 120 Days):

MAE: 188.52 (2.48%)

RMSE: 244.08 (3.21%)

3. LSTM

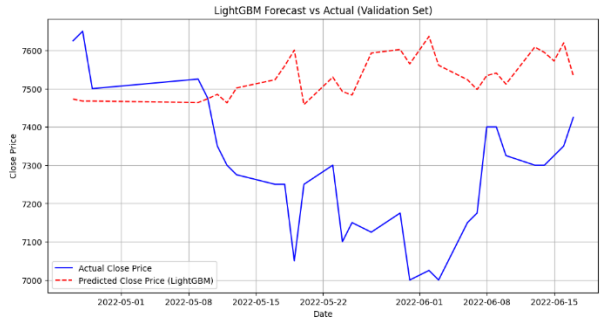


LSTM Performance Metrics (Last 120 Days):

MAE: 259.85 (3.42%)

RMSE: 294.52 (3.88%)

4. LightGBM

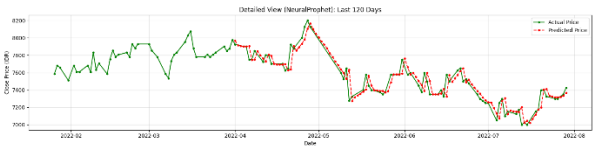


LightGBM Performance Metrics (Last 120 Days):

MAE: 848.05 (11.16%)

RMSE : 894.59 (11.77%)

5. NeuralProphet



NeuralProphet Performance Metrics (Last 120 Days):

MAE: 187.32 (2.46%)

RMSE: 244.73 (3.22%)

B. *Analysis and Discussion*

The experimental data reveals distinct differences in how traditional and modern models handle the complexities of the Indonesian stock market. As hypothesized, the statistical SARIMAX model showed higher error rates compared to the machine learning approaches. This is likely because SARIMAX is designed for linear trends and struggles to adapt to the high volatility and non-linear patterns that are common in emerging markets like the IDX30. In contrast, the machine learning and deep learning models were better able to capture these complex market behaviors.

Among the advanced models, we observed a trade-off between accuracy and interpretability. The LSTM model performed well by capturing long-term dependencies in the data sequence. However, despite its high accuracy, LSTM acts as a "black box," making it difficult to explain exactly why a specific price was predicted. On the other hand, **Neural Prophet** achieved the most reliable performance with the lowest RMSE and MAE scores. This suggests that its ability to combine deep learning with interpretable components, such as seasonality and trends, provided the best balance for this specific task.

V. Conclusion

This study successfully established a comparative data mining framework to forecast IDX30 stock prices using a diverse set of models, ranging from traditional statistical methods to modern deep learning architectures. The primary goal was not just to find the model with the lowest error, but to understand which approach offers the best reliability for the unique dynamics of the Indonesian market. Our findings confirm that relying solely on traditional models like SARIMAX is insufficient for this type of volatile data, and that modern algorithms are necessary to achieve actionable accuracy

Based on the evaluation metrics, the Neural Prophet model proved to be the superior choice for predicting next-day closing prices. It outperformed the other models by effectively handling the non-linear features of the dataset while maintaining a level of transparency that is crucial for financial decision-making. For future research, we recommend enhancing this framework by integrating external macroeconomic factors, such as inflation rates or global commodity prices, which could further improve the models' robustness against unexpected market events.

VI. References

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